Zoidberg 2.0 — Project Summary

Pneumonia Detection from Chest X-Ray Images

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1. PROJECT OVERVIEW

Project Name	Zoidberg 2.0
Institution	Epitech — MSc IT (Machine Learning)
Task	Pneumonia Detection from Chest X-Ray Images
Classification Types	Binary + 3-Class
Dataset Size	Train: 5,216 | Val: 16 | Test: 624
Image Resolution	128×128 grayscale
Preprocessing	StandardScaler + PCA (87.88% variance)

2. BINARY CLASSIFICATION RESULTS (Normal vs Pneumonia)

Split 1: 80/20 stratified split

Model	Accuracy	F1	ROC-AUC
SVM (RBF)	0.9713 ★	0.9805 ★	0.9952 ★
MLP	0.9617	0.9741	0.9937
KNN (K=9)	0.9617	0.9745	0.9809
Logistic Regression	0.9598	0.9728	0.9891
Random Forest	0.9416	0.9617	0.9858

Split 2: Official test set (Dataset 3)

Model	Accuracy	F1	ROC-AUC
CNN	0.8574	0.8947	0.9499
SVM (RBF)	0.7949 ★	0.8562	0.9161
MLP	0.7837	0.8512	0.9032
KNN (K=9)	0.7676	0.8419	0.8741
Logistic Regression	0.7484	0.8310	0.8973
Random Forest	0.7436	0.8287	0.9160

Split 3: 5-Fold Cross-Validation

Model	Accuracy	F1	ROC-AUC
SVM (RBF)	0.9682 ★	0.9784 ★	0.9952 ★
MLP	0.9680	0.9785	0.9928
Logistic Regression	0.9559	0.9704	0.9871

KNN (K=9)	0.9515	0.9677	0.9838
Random Forest	0.9329	0.9562	0.9833

3. 3-CLASS CLASSIFICATION (Normal / Bacteria / Virus)

Model	Split 1 Acc	Split 2 Acc	Split 3 CV Acc
SVM (OvR)	0.8151	0.6939	0.7991
KNN (K=18)	0.7835	0.7019	0.7697
CNN (softmax)	N/A	0.7228	N/A

4. SAVED MODEL ARTIFACTS

Binary Models	Description
svm_best.joblib	SVM + Scaler + PCA pipeline
knn_best.joblib	KNN + Scaler + PCA pipeline
cnn_best.keras	CNN Keras model (64×64 input)
3-Class Models	**Description**
svm3_best.joblib	SVM 3-class + Scaler + PCA
knn3_best.joblib	KNN 3-class + Scaler + PCA
cnn3_best.keras	CNN 3-class Keras model

5. PROJECT COMPLETION STATUS

Item	Status	Details
Dataset	✓ Done	5,216 train + 16 val + 624 test images
Preprocessing	✓ Done	StandardScaler + PCA (100 components)
Binary Models	✓ Done	LR, RF, MLP, SVM, KNN, CNN all trained
3-Class Models	✓ Done	SVM, KNN, CNN for 3-class task
Metrics	✓ Done	Accuracy, F1, ROC-AUC, Precision, Recall
Visualizations	✓ Done	28+ figures (confusion matrices, ROC, comparisons)
Notebook	✓ Done	82 code + 10 markdown cells, fully executable
Model Artifacts	✓ Done	7 saved models (.joblib + .keras)
Documentation	✓ Done	README, HTML/PDF exports, this summary

6. KEY FINDINGS & INSIGHTS

1. **Best Binary Model:** SVM with RBF kernel — 97.13% accuracy on Split 1, 79.49% on official test
2. **CNN Advantage:** Achieves 85.74% on official test — best performance on Split 2 (handles class imbalance well)
3. **Performance Gap:** 97% → 79% drop indicates domain shift between train and test distributions
4. **3-Class Difficulty:** SVM achieves ~80% accuracy on Split 1, 69% on official test (more challenging task)
5. **Model Consistency:** SVM and MLP show best cross-validation results (mean 96.8% and 96.8% accuracy)

7. RECOMMENDATIONS FOR NEXT STEPS

- **Data Augmentation:** Rotations, flips, elastic deformations to improve generalization
- **Transfer Learning:** Use pre-trained CNN (ResNet, DenseNet) on ImageNet
- **Ensemble Methods:** Combine SVM, CNN, KNN with voting or stacking
- **Hyperparameter Tuning:** GridSearch for C, gamma (SVM) and architecture (CNN)
- **Handle Class Imbalance:** SMOTE, oversampling for minority classes
- **Explainability:** SHAP/LIME to interpret model predictions for medical use

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Project: Epitech MSc IT — Machine Learning
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